



# Generation 1 Master Engineering & IP Diligence Baseline

Document HSH-00 | Revision 23 — Professional Master Edition

CONCEPTUAL ENGINEERING BASELINE — PRE-SIMULATION / PRE-HARDWARE / PRE-PROTOTYPE

Proprietary Concept Development Material — Not for Public Disclosure

|                    |                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------|
| Founder            | Amirbahador Atai                                                                              |
| Document Number    | HSH-00                                                                                        |
| Revision           | Rev 23 — Professional Master Edition (supersedes Rev 12–22)                                   |
| Status             | FINAL GENERATION 1 CONCEPTUAL MASTER BASELINE                                                 |
| Development Stage  | Pre-Simulation / Pre-Hardware / Pre-Prototype                                                 |
| Intended Audience  | Technical due-diligence teams, prospective IP licensees, investors, regulatory liaison        |
| Certification Note | Not a certification approval. No design herein is certified or approved by any authority.     |
| Patent Note        | Not a filed patent. IP-A through IP-D are conceptual claim surfaces pending prior-art search. |

**Engineering Integrity Statement.** Every numerical and functional value in this baseline is a TARGET, ASSUMPTION, STUDY, CALC, CONCEPT, or OPEN item unless explicitly marked otherwise. None of these values constitutes demonstrated performance, and nothing in this document shall be read as a certification finding, an airworthiness determination, a granted patent claim, or proof of achieved survival probability. Every status label is defined precisely in Part I, Section 5.

## Document Control

| Field              | Value                                                                               |
|--------------------|-------------------------------------------------------------------------------------|
| Document Title     | HSH Heavy Safe Helicopter — Generation 1 Master Engineering & IP Diligence Baseline |
| Document Number    | HSH-00                                                                              |
| Revision           | Rev 23 — Professional Master Edition                                                |
| Supersedes         | HSH-00 Rev 12 through Rev 22 (all prior presentation and diligence packages)        |
| Status             | FINAL G1 CONCEPTUAL MASTER BASELINE                                                 |
| Founder            | Amirbahador Atai                                                                    |
| Development Stage  | Pre-Simulation / Pre-Hardware / Pre-Prototype                                       |
| Classification     | Proprietary Concept Development Material — Not for Public Disclosure                |
| Certification Note | Not a certification approval                                                        |
| Patent Note        | Not a filed patent                                                                  |

■ *Editorial scope of Rev 23. This edition is a presentation, structure, and consistency upgrade of the Rev 22 FULL technical baseline. It reorganizes the material into nine parts with a working table of contents and an executive summary, standardizes terminology and units, corrects internal cross-references, and improves table formatting for reviewer usability. It does not add new engineering analysis, new test evidence, or new closure of any open item, and it does not upgrade any TRL, status label, or claim beyond what Rev 22 FULL supported. Where this edition's wording differs from Rev 22 FULL, the intent is clarity, not a change of technical content; the Rev 22 values remain the controlling baseline for configuration purposes.*

## Revision History

| Rev       | Primary Change                                                                                                                                              | Status                                 |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Rev 12    | 26,000 kg / 26.0 m / 4.0 m baseline; 8-canopy recovery; single 6 s target; 10–12 occupants                                                                  | <b>Superseded</b>                      |
| Rev 13    | Main-rotor separation removed from core; staged S0–S5; 3+2 canopies; Risk Register; Prior Art                                                               | <b>Superseded</b>                      |
| Rev 14–16 | Full narrative depth; helium-hybrid trade posture; competitive IP narrowing                                                                                 | <b>Superseded</b>                      |
| Rev 17–18 | HSH identity unified; 8-person lock; commercial content separated; Maturity Framework                                                                       | <b>Superseded</b>                      |
| Rev 19    | 16-airbag architecture; seat specification; OI-16 / OI-17 registered                                                                                        | <b>Superseded</b>                      |
| Rev 20    | Seat stroke $\geq 150$ mm / $\leq 6,670$ N; airbag mass 200 kg; OI-18; IP-D; TRL; competitive; defined-vs-open; geometry; mission                           | <b>Superseded</b>                      |
| Rev 21–22 | Diligence §§32–41; portrait uniformity; dense no-blank composition of full content                                                                          | <b>Superseded as presentation base</b> |
| Rev 23    | Professional master edition: nine-part structure, working TOC, executive summary, standardized tables/units, consolidated glossary, reviewer-usability pass | <b>CURRENT</b>                         |

## Baseline Retention Statement

Every requirement, status label, open item (OI-01 through OI-18), risk register entry, verification gate, interface definition, safety layer, Safety Kernel precondition set, staged separation timeline, recovery architecture, prior-art posture, and engineering maturity framework carried in Rev 18 through Rev 22 is retained in Rev 23 without technical alteration. Only presentation, structure, and cross-reference quality are improved. No open item is presented as closed, and no status label is upgraded, in this edition.

## Executive Summary

HSH Heavy Safe Helicopter (Generation 1) is a conceptual architecture for a 26-ton-class single-main-rotor helicopter organized around one idea: an occupied survival capsule that can become the aircraft's final landing boundary in a catastrophic emergency, supported by a deterministic safety-authorization layer, a staged non-rotor separation sequence, an eight-point capsule-integrated impact-attenuation system, and a distributed parachute recovery system. The programme is positioned as a distinct product category — a survivability-first rotorcraft — rather than a payload or cabin-space competitor to existing helicopter models.

**What this baseline is.** HSH-00 Rev 23 is a Generation 1 conceptual master baseline: a single-source document intended to let a technical due-diligence team, prospective IP licensee, or investor evaluate the architecture without reconstructing it from an underlying tree of hundreds of subordinate documents. It defines top-level geometry and mass allocations, a six-layer survival architecture, a Safety Kernel precondition set, a staged S0–S5 separation timeline, a 3+2 canopy recovery architecture, an eight-support landing system, a 16-airbag and crashworthy-seat occupant-protection architecture, four candidate IP claims (IP-A through IP-D), a full open-items register (OI-01 through OI-18), a technology-readiness assessment, a competitive-differentiation table, and an evidence-ladder framework connecting every claim to the test and analysis evidence required before it can be treated as demonstrated.

**What this baseline is not.** No element of the design has been analyzed by simulation, demonstrated by test, validated in flight, or certified by a regulator. Every numerical target is a planning allocation, not a verified result. The document deliberately excludes main-rotor separation from the core safety case — a design decision made in Rev 13 and retained here — because no credible six-degree-of-freedom analysis or structural evidence exists to support separating a rotor carrying several hundred megajoules of rotational kinetic energy from a descending occupied capsule. This exclusion is treated as a strength of the architecture, not a gap: a safety case should not depend on its least-proven capability.

| Dimension                       | Current Position                                                                                                                                                                                                                                    |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core proposition</b>         | Occupied survival capsule as final landing boundary, gated by a deterministic Safety Kernel, supported by staged non-rotor separation, eight-point impact attenuation, and 3+2 canopy recovery.                                                     |
| <b>Maturity</b>                 | <b>Specified / Studied</b> across all subsystems. TRL 1–4 by subsystem (Part V). No subsystem has reached subsystem testing (E3) or beyond.                                                                                                         |
| <b>Principal residual risks</b> | Stage S3 capsule release has no certified analog; minimum canopy deployment altitude unresolved; 7,000 kg capsule mass unverified bottom-up; sealed-capsule airbag overpressure/toxicity (OI-18) unanalyzed; six-layer independence undemonstrated. |
| <b>IP posture</b>               | Four candidate claims (IP-A–D), strongest as a system-level combination (IP-D). Conceptual only — not filed; professional prior-art search required before any filing or broad non-NDA disclosure.                                                  |
| <b>Locked configuration</b>     | 8-occupant maximum (Generation 1 hard limit); main-rotor separation excluded from the core safety case; no retractable landing gear; central capsule position.                                                                                      |
| <b>Evidence path</b>            | G0 architecture freeze (approaching closure) through G8 certification/qualification, structured against the E0–E5 evidence ladder (Part VII).                                                                                                       |

■ This executive summary is new to Rev 23 and is provided for reviewer convenience. It synthesizes, but does not alter, the technical content of Parts I through IX. Where a summary statement and a detailed section appear to differ, the detailed section is controlling.

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## PART I

# Programme Baseline, Scope and Status Control

Identity, occupancy lock, scope, philosophy, status-label definitions and configuration-change triggers

## 1. Baseline Freeze and Controlled Identity

### 1.1 Programme Identity

The programme name is HSH Heavy Safe Helicopter. The controlling document is HSH-00 Revision 23, Founder: Amirbahador Atai. All prior references to HSC-00 or earlier HSH-00 revisions are superseded for configuration control. Rev 23 carries the complete Rev 20/22 technical baseline in a reorganized, consistency-checked presentation.

### 1.2 Occupancy Lock

■ **8 persons maximum [LOCKED LIMIT]. This is a Generation 1 hard configuration constraint.**

Any future change requires formal configuration control and re-analysis of geometry, mass, CG, inertia, seating, restraints, airbag count, life support, egress, structural loads, recovery sizing and certification basis.

### 1.3 What Rev 20/22 Changed Relative to Rev 19

| Change                                                   | Rev 19                    | Rev 20/22/23                                    | Rationale                                                        |
|----------------------------------------------------------|---------------------------|-------------------------------------------------|------------------------------------------------------------------|
| Seat specification                                       | ≥50 cm ergonomic foam     | ≥150 mm dynamic stroke;<br>≤6,670 N spinal load | Foam thickness is not an aviation crashworthy-seat criterion     |
| Airbag mass budget                                       | 80–120 kg                 | 200 kg                                          | Automotive benchmark scaled to 16 aviation-grade units           |
| Sealed-capsule airbag hazard                             | Not explicitly registered | OI-18 added                                     | 16 airbags in sealed volume: overpressure and gas-toxicity risk  |
| IP-D claim                                               | Not present               | Added                                           | System-level combination claim stronger than any single IP-A/B/C |
| TRL / competitive / defined-vs-open / geometry / mission | Not present               | Added                                           | Required for serious technical due-diligence                     |

### 1.4 What Is Retained Unchanged

Every requirement, status label, open item OI-01 through OI-18, risk register entry, verification gate, interface definition, safety layer, Safety Kernel precondition set, staged separation timeline, recovery architecture, prior-art posture, and engineering maturity framework from Rev 18/19/20/22 is retained. Rev 23 corrects only structure, cross-references and presentation.

## 2. Purpose and Scope

### 2.1 Programme Objective

HSH Generation 1 defines the system-level architecture for a 26-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain — not reliance on any single device.

### 2.2 Core Scoping Decision

The survival chain is designed to work assuming the main rotor is never separated from the aircraft. This deliberate decision was introduced in Rev 13 and is retained. No credible trajectory, gyroscopic, or structural analysis yet exists to support main-rotor separation from a descending occupied capsule. Building the core safety case on an unproven capability would undermine confidence in the entire architecture and in all primary IP claims.

### 2.3 Scope Coverage

This document covers: the aircraft; the occupied survival capsule; the rapid selective separation system (non-rotor components only); safety initiation paths; deterministic safety authorization; stabilization; recovery; impact attenuation; post-event survival; and verification interfaces. It does not cover detailed structural analysis results, certified engine data, qualification test results, airworthiness findings, or IP legal opinions — those are evidence-gate deliverables.

### 2.4 Design Philosophy

Separation is a controlled emergency mode, authorized only when defined conditions are satisfied. The survival architecture supports multiple outcomes in priority order: (1) conventional flight recovery; (2) autorotation; (3) selective hazard/mass reduction; (4) capsule separation and parachute recovery; (5) energy-absorbing landing; (6) post-landing rescue. The most extreme mode shall not be required for manageable failures.

### 2.5 Human Authority

Human emergency authority is preserved and non-bypassable. No AI system, optimization algorithm, or predictive model shall override the crew's ability to command, inhibit, or terminate an automatic survival sequence. This is a design requirement, not a guideline.

## 3. What Is Defined Today vs. What Remains Open

■ This section is the first thing a technical due-diligence team should read.

| Domain              | What Is Defined                                                                                      | What Remains Open                                           | Gate  |
|---------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------|
| System architecture | Six-layer survival; Safety Kernel; state machine; staged separation; 3+2 recovery; 8-support landing | All engineering analysis                                    | G1    |
| Mass accounting     | 26,000 kg and 7,000 kg planning models; component-level allocation                                   | Bottom-up verification; uncertainty and growth              | G1    |
| Capsule structure   | CFRP roll-cage concept; titanium nodes; load-path description                                        | FEA; crash-load analysis; fatigue                           | G5    |
| Separation system   | S0–S5 timeline; candidate non-rotor components; mandatory protections                                | S3 mechanism; 6-DOF; ground demonstration                   | G2–G3 |
| Recovery system     | 3+2 canopy architecture; sizing rationale; failure-tolerance concept                                 | Canopy sizing; inflation dynamics; min. deployment altitude | G6    |
| Landing supports    | 8-point concept; energy absorber; asymmetric-failure requirement                                     | FEA; drop tests; terrain interaction                        | G5    |

| Domain              | What Is Defined                                                       | What Remains Open                                                  | Gate                |
|---------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|---------------------|
| Crashworthy seats   | ≥150 mm stroke; ≤6,670 N criterion; 8 positions                       | Seat design; foam/stroking-leg; dynamic tests                      | G2–G5               |
| Airbag system       | 16-bag concept; 2 per occupant; placement; 200 kg budget              | Packaging; deployment logic; sealed-volume gas (OI-18); sled tests | G2–G5               |
| Safety Kernel       | Precondition set defined; verification requirements stated            | Software/hardware implementation; formal verification              | G1–G2               |
| IP claims           | IP-A, IP-B, IP-C, IP-D conceptually defined; prior-art posture stated | Professional prior-art search; claim drafting; filing              | External legal      |
| Certification basis | CS-29 referenced                                                      | Formal regulatory engagement; means of compliance                  | External regulatory |
| Performance         | Hover power study (3.57 MW); disk-loading benchmark                   | All performance analysis; cruise; OEI; range closure               | G1                  |
| Mission profile     | 1,000 km range target; 8 occupants                                    | Fuel model; cruise speed; altitude; reserve policy                 | G1                  |

## 4. Aircraft Geometry and Configuration

### 4.1 Rotor Disk Loading — Benchmark Comparison

| Aircraft                             | MTOW             | Main Rotor Diameter | Disk Loading                   |
|--------------------------------------|------------------|---------------------|--------------------------------|
| Mi-26 (Russia — heaviest production) | 56,000 kg        | 32.0 m              | ~55 kg/m <sup>2</sup>          |
| CH-53K King Stallion (US heavy-lift) | 38,400 kg        | 24.1 m              | ~58–60 kg/m <sup>2</sup>       |
| CH-47F Chinook (tandem, per disk)    | 22,680 kg        | 18.3 m per rotor    | ~43 kg/m <sup>2</sup> per disk |
| Sikorsky S-92 (medium offshore)      | 12,020 kg        | 17.17 m             | ~52 kg/m <sup>2</sup>          |
| AW139 (medium — market neighbor)     | 6,800 kg         | 13.8 m              | ~45 kg/m <sup>2</sup>          |
| <b>HSH Gen1 Rev23</b>                | <b>26,000 kg</b> | <b>26.0 m</b>       | <b>48.97 kg/m<sup>2</sup></b>  |

Design rationale: HSH's disk loading is deliberately lower than the Mi-26 and CH-53K. This trades rotor size for: (a) greater hover-power margin — needed given the additional mass of the capsule, separation system, airbags and crashworthy seats; (b) reduced downwash velocity at the eight fixed landing supports; and (c) lower tip-speed requirement, reducing rotor noise. To be confirmed or revised at Document 063 closure.

## 4.2 Aircraft Geometry Summary

| Geometry Item                   | Preliminary Definition                                      | Status            |
|---------------------------------|-------------------------------------------------------------|-------------------|
| Overall length                  | Approx. 16.5–18.0 m                                         | TARGET/STUDY      |
| Maximum fuselage width          | Approx. 4.0–4.5 m                                           | TARGET/STUDY      |
| Overall height                  | Approx. 4.8–5.3 m                                           | TARGET/STUDY      |
| Main rotor diameter             | 26.0 m                                                      | TARGET            |
| Tail rotor diameter             | 4.0 m                                                       | TARGET            |
| Rotor arrangement               | Single main rotor + tail rotor                              | REQ/STUDY         |
| Landing supports                | 8 fixed, short, mechanical supports under capsule           | TARGET            |
| Retractable landing gear        | None                                                        | LOCKED<br>CONCEPT |
| Capsule position                | Central, integrated within primary airframe                 | LOCKED<br>CONCEPT |
| Capsule normal-flight load path | Primary attachment system with controlled emergency release | REQ               |
| Capsule emergency load path     | Independent final landing element after verified clearance  | REQ               |

## 5. Baseline Status and Configuration Control

### 5.1 Status Label Definitions

| Status Label                         | Precise Meaning                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------|
| [LOCKED CONCEPT]                     | Retained as baseline design principle; not to be changed without formal impact assessment  |
| [LOCKED LIMIT]                       | Hard Generation 1 constraint; requires formal change process and full re-analysis to alter |
| [REQ]                                | Top-level requirement; binding on the design unless formally waived                        |
| [TARGET]                             | Numerical or functional development objective; not demonstrated                            |
| [ASSUMPTION / STUDY]                 | Working engineering assumption for preliminary sizing; to be replaced by analysis or test  |
| [CALC]                               | Calculated from stated assumptions; inputs are themselves assumptions or targets           |
| [OPEN]                               | Requires analysis, simulation, test, supplier data or design decision                      |
| [OPEN — CRITICAL]                    | Open item with material effect on the core safety case; priority closure required          |
| [DEFERRED — RESEARCH TRACK]          | Excluded from licensable core baseline; not to be presented as core product                |
| [TBD — DEPENDENCY]                   | Cannot be fixed until a dependent activity closes                                          |
| [PRIOR ART STATUS: NOT YET SEARCHED] | No novelty conclusion implied; professional search required                                |

### 5.2 Configuration Change Triggers

A formal configuration change is required for any of the following: MTOW; main rotor diameter; tail rotor diameter; capsule mass; occupant count; separation-sequence architecture; parachute architecture; landing-support count or geometry; airbag count or placement; seat-stroke specification; Safety Kernel precondition set; any safety-layer definition. Each change triggers an impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

PART II

Safety and Survival Architecture

Six-layer survival chain, Safety Kernel, separation system, capsule, landing supports, airbags, seats, recovery canopies, core safety case, timeline, state machine, failure modes, interfaces

6. Six-Layer Survival and Safety Architecture

The six layers are functionally distinct and independently capable — not merely redundant. Independence means physically, electrically, and software-separate implementations that cannot both be defeated by any single common-cause failure.

| Layer | Function                                             | Primary Purpose                                                                                | Independence Requirement                                                      |
|-------|------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| L1    | Primary automatic emergency detection and initiation | Detect catastrophic states; initiate validated survival sequence when Safety Kernel authorizes | Separate processor, sensor suite, power bus from L2                           |
| L2    | Independent backup initiation                        | Second segregated automatic path if L1 unavailable or in disagreement                          | Different physical location, software architecture, power source from L1      |
| L3    | Independent manual/crew authority                    | Crew initiation, inhibit and emergency authority without L1/L2                                 | Hardwired path bypassing L1/L2; software cannot disable                       |
| L4    | Rapid selective physical separation                  | Release designated non-rotor hazards using controlled mechanisms                               | Dual independent actuation paths; no single-point failure defeats both        |
| L5    | Direct-impact survival                               | Protect occupants via capsule structure, restraints, seats and eight energy-absorbing supports | Passive structural system; no active components required for primary function |
| L6    | Recovery and post-event survival                     | Deploy recovery canopies; locate occupants; support rescue                                     | Independent emergency power; independent comms; independent locator beacon    |

■ **AI Boundary — Non-Negotiable Principle: AI or advanced prediction may assist detection, classification and decision support. Deterministic safety logic authorizes safety-critical actuation. AI confidence is never equivalent to safety authorization.**

This principle is non-negotiable from a certification perspective and is the engineering basis for IP-A (Part III, Section 17).

Survival Chain

PREVENT/MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE (NON-ROTOR) → VERIFY CAPSULE CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT (8 supports + crashworthy seats + 16 airbags) → SURVIVE / LOCATE / RESCUE → AUDIT

7. Deterministic Safety Authorization and Safety Kernel

The Safety Kernel prevents AI, digital-twin models, and predictive systems from directly commanding hazardous separation without satisfying a defined set of physically verifiable preconditions. It is the boundary between the decision layer and the actuation layer.

## 7.2 Safety Kernel Precondition Set

| Precondition Category      | Required Evaluation                                                                                                                             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Altitude and airspeed      | Above minimum recovery altitude; within separation dynamic-pressure envelope; within parachute deployment envelope accounting for sequence time |
| Attitude and angular rates | Pitch, roll, yaw within acceptable separation envelope; angular rates within acceptable separation rate limits                                  |
| Rotor and engine state     | Rotor RPM documented; engine/fire/fuel hazard state known and consistent with separation sequence                                               |
| Structural and attachment  | Capsule attachment health confirms release mechanisms functional; structural health consistent with planned separation                          |
| Trajectory and clearance   | Acceptable clearance between released components and capsule/parachutes confirmed (real-time or pre-computed envelope)                          |
| Recovery system            | Parachute deployment envelope satisfiable after separation; recovery integrity confirmed                                                        |
| Power and communication    | Independent survival power available; communication loss does not inhibit Safety Kernel                                                         |
| Human authority            | Manual crew inhibit confirmed; rollback/fallback availability assessed                                                                          |
| Airbag authorization       | Impact state valid; deployment permitted under deterministic preconditions; capsule-interior environment suitable                               |

## 7.3 Kernel Decision Table

| Decision             | Minimum Design Question                                         | Verification Method                            |
|----------------------|-----------------------------------------------------------------|------------------------------------------------|
| Authorize separation | Are all mandatory preconditions simultaneously true?            | Formal logic review + fault injection          |
| Inhibit separation   | Can an unsafe state block actuation even under software fault?  | Fault injection + power-loss test              |
| Fallback mode        | What happens if a sensor or command path is lost?               | Failure-mode testing per FMEA                  |
| Human override       | Is authority bounded, deterministic and physically independent? | HMI + system integration test                  |
| Airbag authorize     | Is impact state valid and deployment safe in closed capsule?    | Logic review + sled/impact test + gas analysis |
| Audit record         | Can every emergency event be fully reconstructed?               | Data-recording verification test               |

The Safety Kernel shall be implemented with: formally verified software or hardware logic; documented requirement set with code traceability; independence from AI/ML layers; segregated power supply; independent communication path to actuation hardware; protection against command injection; and an immutable audit record. No update or in-flight reprogramming shall modify the Safety Kernel precondition set without a formal change process and re-verification.

## 8. Controlled Rapid Separation System

The system is selective, staged, and controlled — not a blanket jettison of everything. State-dependent release logic means the set of components released in a fire emergency differs from the set in a structural emergency. Every component designated as separable must be confirmed through structural, aerodynamic, dynamic, fire, control, trajectory and collision analysis.

## 8.2 Candidate Separable Components

| Component                              | Condition for Separation                                 | Analysis Requirement                              |
|----------------------------------------|----------------------------------------------------------|---------------------------------------------------|
| Main engines (both)                    | After fuel isolation; where structurally separable       | Trajectory, structural, thermal, fire analysis    |
| Selected transmission/drive components | Where release is structurally and dynamically acceptable | 6-DOF dynamics, CG-shift analysis                 |
| Tail rotor assembly + gearbox + drive  | Where aerodynamically and dynamically safe               | Anti-torque state, trajectory, collision analysis |
| Primary fuel tanks + major fuel lines  | After confirmed fuel isolation                           | Fire-hazard analysis, isolation timing            |
| Selected external structural panels    | High-mass, not on capsule primary load path              | Structural, trajectory, collision analysis        |
| Main rotor blades and hub              | <b>EXCLUDED from core candidate list</b>                 | Deferred research track — §12                     |

## 8.3 Mandatory Release Protections

| Protection                           | Requirement                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------|
| Positive mechanical retention        | Active positive lock during all normal flight; release requires deliberate actuation     |
| Electrical/logic inhibit             | Release electrically inhibited in normal flight state; specific arming sequence required |
| Independent command paths            | Primary and backup paths physically and electrically independent                         |
| Defined arm/fire sequence            | Documented, tested, protected against out-of-sequence commands                           |
| Release-state sensing                | Positive confirmation of attached/released state at each stage                           |
| Fuel/energy isolation                | Isolation precedes or is concurrent with release; no new uncontrolled hazard created     |
| Asymmetric/partial-release detection | Full reporting of any case where less than the designated set is released                |
| Post-release monitoring              | Track released components where practically possible                                     |
| Capsule-clear verification           | Capsule release not authorized until all non-rotor components confirmed clear            |
| Failed-release fallback              | Safe state defined for every possible failed-release configuration                       |

## 8.4 Separation System Mass Budget — 1,000 kg

| Separation System Component                         | Mass            |
|-----------------------------------------------------|-----------------|
| Primary capsule release mechanisms and locks        | 240 kg          |
| Independent backup release mechanisms               | 150 kg          |
| Local structural reinforcement and release fittings | 160 kg          |
| Independent initiators / actuators                  | 90 kg           |
| Release-state sensors and monitoring                | 65 kg           |
| Independent power / signalling and protected wiring | 70 kg           |
| Fuel / energy isolation interfaces                  | 55 kg           |
| Retention, inhibit and safety-interlock hardware    | 50 kg           |
| Configuration growth allowance + integration margin | 120 kg          |
| <b>TOTAL</b>                                        | <b>1,000 kg</b> |

## 9. Occupied Survival Capsule

The capsule is not a cabin with a parachute. It is a structurally independent survival system that functions as the passenger compartment during normal flight and becomes the final landing boundary in a catastrophic emergency. Every subsystem inside the capsule must function independently of the main airframe after separation. Primary structure: continuous CFRP roll-cage with metallic/titanium nodes at load-introduction points. CFRP provides high specific strength; titanium nodes provide fatigue-resistant, corrosion-resistant, impact-resistant interfaces at maximum stress-concentration points. Continuous roll-cage geometry provides controlled deformation paths — progressive outward deformation rather than inward collapse.

| Capsule Parameter            | Rev 23 Definition                                             | Status       |
|------------------------------|---------------------------------------------------------------|--------------|
| Occupancy                    | 8 persons maximum                                             | LOCKED LIMIT |
| Complete capsule mass        | 7,000 kg planning target                                      | TARGET       |
| Primary structure            | Continuous CFRP roll-cage, titanium nodes                     | CONCEPT      |
| Seat type                    | Energy-absorbing, ≥150 mm stroke, ≤6,670 N peak spinal load   | TARGET/STUDY |
| Airbag system                | 16 total (2 per occupant)                                     | TARGET/STUDY |
| Landing supports             | 8 fixed, progressive energy absorbers                         | TARGET       |
| Recovery canopy architecture | 3 primary + 2 reserve                                         | STUDY        |
| Independent emergency power  | Battery-based, independent of airframe bus                    | TARGET       |
| Life support                 | Emergency oxygen, minimum post-landing endurance              | TARGET       |
| Capsule-clear verification   | Required before recovery deployment                           | REQ          |
| Post-landing stability       | 8-point architecture; asymmetric/failed-leg analysis required | REQ          |
| Water landing                | Flotation, sealing for ditching scenario                      | TARGET       |
| Emergency egress             | Multiple exits rated for crash loads; airbag-compatible paths | TARGET       |

## 10. Eight Fixed Short Mechanical Landing Supports

Eight short, thick, fixed mechanical supports are integrated with the capsule lower structure. They are not conventional helicopter landing gear. Key rationale: **capsule-integrated** — supports remain with occupants after separation; **redundancy** — loss of one or two supports still leaves six or seven functional; **energy absorption** — each support incorporates a progressive mechanical energy-absorber with controlled, bounded crush force throughout deformation without a hard metallic stop.

| Impact Velocity | Total KE (7,000 kg capsule) | KE per Support (÷8) |
|-----------------|-----------------------------|---------------------|
| 6 m/s           | 126.0 kJ                    | 15.75 kJ            |
| 8 m/s           | 224.0 kJ                    | 28.0 kJ             |
| 10 m/s          | 350.0 kJ                    | 43.75 kJ            |

■ These are sizing illustrations, not allowable impact energies. Actual survival depends on pulse shape, attitude, angular rates, terrain and structural response. Seat stroke and airbags provide additional occupant load management after the supports arrest capsule motion.

| Case                         | Analysis Focus                              | Required Evidence  |
|------------------------------|---------------------------------------------|--------------------|
| All 8 supports, flat terrain | Load sharing and stability                  | FEA + drop test    |
| 1 support failed             | Redistribution and local overload           | FEA + test         |
| 2–3 supports failed          | Survivability boundary                      | Sensitivity study  |
| Sloped terrain (≤15°)        | Roll/tip tendency, asymmetric load transfer | Terrain simulation |

| Case                    | Analysis Focus                     | Required Evidence        |
|-------------------------|------------------------------------|--------------------------|
| Soft ground / snow      | Penetration and energy dissipation | Ground interaction test  |
| Water impact (ditching) | Hull sealing and flotation         | Water impact test        |
| Oblique combined loads  | Lateral + vertical + longitudinal  | 6-DOF dynamic simulation |

10A. Sixteen-Airbag Occupant Protection Architecture

| Item                      | Rev 23 Baseline                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------|
| Occupant positions        | 8 (locked maximum)                                                                          |
| Airbags per occupant      | 2                                                                                           |
| Total airbags             | 16                                                                                          |
| Primary set               | 8 — one per seat, head/torso / principal impact direction                                   |
| Secondary set             | 8 — one per seat, lateral / lower-body / secondary direction                                |
| Status                    | TARGET/STUDY                                                                                |
| Mass planning (corrected) | ≈ 200 kg (16 aviation-grade modules + inflators + ECU + pyrotechnic harness + mounting)     |
| Verification              | Sled tests, inflation tests, interference checks, gas-toxicity review, egress compatibility |

**Placement principles.** Eight primary + eight secondary; shall not obstruct egress; compatible with CFRP roll-cage, seats and restraints; deployment only under deterministic Safety Kernel preconditions; airbag authorization passes through the Safety Kernel — not an independent system that can deploy outside the authorization chain.

**Mass correction rationale.** Rev 19 estimated 80–120 kg. Aviation-grade modules at 1.5–2.0 kg each × 16 ≈ 24–32 kg module mass. Adding ECU (8–15 kg), pyrotechnic harness (20–30 kg), mounting (30–40 kg), wiring (15–20 kg), and qualification margin yields approximately 150–220 kg, rounded to 200 kg. The Rev 19 estimate was insufficient and would have created a false closure of the capsule mass budget.

■ **Sealed-Capsule Deployment Hazard — OI-18 (CRITICAL).** Sixteen airbags deploying simultaneously in a sealed capsule of limited internal volume create risks not present in automotive applications: (a) overpressure — rapid gas generation may transiently exceed safe ear-drum and sinus pressure limits; (b) gas composition — propellant gases dilute capsule oxygen; (c) temperature — inflator heat in a sealed volume; (d) particulate — sodium compounds from certain propellant chemistries. Mitigation options include staggered sequential deployment, vented inflators, or reduced inflator energy for the closed-capsule environment. Must be analyzed before deployment logic is locked.

10B. Eight Crashworthy Seats — Corrected Specification

Rev 19 specified “≥50 cm ergonomic foam seating-surface thickness.” This is not a recognized aviation crashworthy-seat criterion. FAR 27.562 (and the CS-29 equivalent) evaluates seats by occupant injury criteria, not foam thickness. Soft foam can compress fully under crash loads in milliseconds, providing minimal protection after initial compliance. The correct engineering specification is a minimum dynamic stroke and a peak load limit.

| Item                               | Rev 23 Baseline                                                                    |
|------------------------------------|------------------------------------------------------------------------------------|
| Number of seats                    | 8                                                                                  |
| Minimum dynamic stroke             | ≥ 150 mm (combined seat/absorber/floor deformation path)                           |
| Peak spinal compressive load limit | ≤ 6,670 N (FAR 27.562 reference criterion, proxy)                                  |
| Force-displacement response        | Progressive — bounded crush force throughout stroke, no hard stop at stroke limit  |
| Seat architecture                  | Energy-absorbing leg + foam/elastomer cushion combination [OPEN — detailed design] |

| Item         | Rev 23 Baseline                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Function     | Limit peak loads transmitted to pelvis, spine and lower extremities during hard landing; works with 8 landing supports + 16 airbags + 5-point restraints + roll-cage |
| Status       | <b>TARGET/STUDY</b> — density curve, stroke profile, fire rating, mass, dynamic performance <b>[OPEN]</b>                                                            |
| Mass         | Included in 700 kg seats + restraints capsule line                                                                                                                   |
| Verification | Dynamic seat tests per FAR 27.562 or CS-29 equivalent; combined drop/sled cases                                                                                      |

11. Recovery Architecture — 3 Primary + 2 Reserve Canopies

Three large primary canopies are precededented by crewed spacecraft recovery systems at comparable mass. Fewer, larger canopies reduce simultaneous-inflation complexity and inter-canopy collision risk at low altitude — the failure mode most common in real rotorcraft emergencies.

| Recovery Mode                       | Use Case                                                    | Status       |
|-------------------------------------|-------------------------------------------------------------|--------------|
| Controlled flight recovery          | Recoverable disturbance                                     | REQ/STUDY    |
| Autorotation                        | Engine/power loss when altitude/rotor state permit          | REQ/STUDY    |
| Selective mass/hazard separation    | Fire, structural or combined emergency                      | REQ/CONCEPT  |
| Capsule separation                  | Catastrophic state                                          | REQ/CONCEPT  |
| 3 primary recovery canopies         | Primary post-separation vertical recovery                   | STUDY        |
| 2 reserve canopies                  | On detected primary failure only                            | STUDY        |
| 8 energy-absorbing landing supports | Final ground-impact attenuation                             | REQ/TARGET   |
| Crashworthy seats + 16 airbags      | Occupant load management after ground contact               | TARGET/STUDY |
| Post-landing survival               | Locator, comms, lighting, emergency power, rescue interface | REQ          |

For a 7,000 kg capsule at a 7.5 m/s target descent rate: total canopy area  $A_{total} \approx 2,983 \text{ m}^2$ ; per canopy ( $\div 3$ )  $\approx 994 \text{ m}^2$  each; corresponding diameter  $\approx 35\text{--}36 \text{ m}$  per primary canopy. Consistent with spacecraft precedent, including Soyuz-class multi-main canopy recovery. Detailed sizing is deferred to Document 159.

■ The minimum deployment altitude is **[OPEN — CRITICAL]** (OI-02). Most rotorcraft accidents occur at low altitude.

12. Core Safety Case — Rebuilt Without Main-Rotor Separation

The single largest credibility risk in Rev 12 was the implicit dependency of the safety case on main-rotor separation — a capability for which no credible 6-DOF simulation, structural trajectory analysis, or physical evidence exists. A 26 m rotor carries several hundred megajoules of rotational kinetic energy and a gyroscopic moment resisting rapid attitude change. Removing this dependency from the core safety case is what makes the remaining architecture defensible.

The survival chain works assuming the main rotor is never separated.

- 1
- Controlled flight recovery / autorotation — no separation required.
- 2
- Selective non-rotor hazard separation — engines, fuel tanks (post-isolation), tail rotor/drive, fire-load panels.
- 3
- Capsule release from airframe — capsule departs beneath/behind the still-attached rotor disk.
- 4
- Capsule descent under the 3+2 canopy system.
- 5
- Impact attenuation via eight fixed supports + crashworthy seats + 16 airbags.
- 6
- Post-landing survival — independent power, life support, locator, communications.

Main-rotor separation is retained as Document 080 deferred research, explicitly excluded from the licensable core baseline and from primary IP claims. If independent 6-DOF analysis and mechanism-test evidence later demonstrate safe separation, it may be reintroduced through the configuration-change process.

13. Staged Separation Timeline

| Stage | Action                                                             | Target Window                | Confidence Basis                    | Status          |
|-------|--------------------------------------------------------------------|------------------------------|-------------------------------------|-----------------|
| S0    | Validate emergency state; arm path; isolate fuel/electrical/energy | 0.0–0.5 s                    | Comparable to ejection-seat arming  | TARGET          |
| S1    | Fuel isolation and designated fire-hazard separation               | 0.5–1.8 s                    | Comparable to crash fuel-shutoff    | TARGET          |
| S2    | Engine + tail-rotor drive separation (non-main-rotor)              | 1.8–3.5 s                    | Comparable to cargo-release, scaled | TARGET/OPEN     |
| S3    | Capsule-to-airframe primary structure release                      | 3.5–5.0 s                    | No direct certified analog          | OPEN — CRITICAL |
| S4    | Capsule clearance verification                                     | 5.0–5.8 s                    | Sensor-response-limited             | TARGET          |
| S5    | Recovery canopy authorization                                      | After S4, gated on clearance | Not time-gated; correctness first   | REQ             |

■ If the architecture cannot safely achieve these windows, targets shall be revised under configuration control — not achieved by reducing verification rigor. Partial and asymmetric separation are first-class emergency states at every stage.

## 14. Emergency State Machine

| State                      | Entry Condition                | Primary Action                                               | Exit / Next                          |
|----------------------------|--------------------------------|--------------------------------------------------------------|--------------------------------------|
| 1 — NORMAL FLIGHT          | All systems nominal            | Continuous monitoring; locks engaged                         | Anomaly → 2                          |
| 2 — ANOMALY DETECTED       | Sensor/model flags abnormality | Increase monitoring; begin confirmation                      | Confirmed → 3; Rejected → 1          |
| 3 — CONFIRMATION           | Potential condition detected   | Independent evidence and validity checks                     | Confirmed → 4                        |
| 4 — IMPACT ASSESSMENT      | Condition confirmed            | Estimate authority, altitude, airspeed, fire/structural risk | Complete → 5                         |
| 5 — RECOVERY ATTEMPT       | Assessment complete            | Conventional control, OEI, autorotation                      | Succeeds → 1; Fails → 6              |
| 6 — SURVIVAL AUTHORIZATION | Recovery failed                | Safety Kernel evaluates all preconditions                    | Authorized → 7; Not met → degraded   |
| 7 — STAGED SEPARATION      | Authorized                     | Execute designated release set (non-rotor)                   | Complete → 8                         |
| 8 — CAPSULE CLEARANCE      | Separation executed            | Verify clearance; identify residual attachments              | Clear → 9; Partial → partial-release |
| 9 — RECOVERY DEPLOYMENT    | Clearance verified             | Authorize parachute within validated envelope                | Deployment confirmed → 10            |
| 10 — DESCENT MANAGEMENT    | Canopies deployed              | Monitor attitude, rates, canopy state, support readiness     | Ground imminent → 11                 |
| 11 — LANDING ATTENUATION   | Ground contact                 | Supports + seats + airbags all load simultaneously           | Landing complete → 12                |
| 12 — POST-LANDING SURVIVAL | Stabilized                     | Locator active; comms active; life support active            | Rescue confirmed → 13                |
| 13 — AUDIT                 | Rescue complete                | Preserve all event data, sensor states, command history      | Data secured — END                   |

## 15. Independence, Failure Modes and Common-Cause Control

| Failure / Hazard                                     | Severity     | Required Design Response                                                                                 |
|------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------|
| Unintended capsule separation in normal flight       | Catastrophic | Mechanical locks + electrical inhibits + command validation + monitoring + quantitative safety objective |
| Partial separation (incomplete component release)    | Critical     | State sensing + asymmetric-load analysis + defined degraded recovery mode                                |
| Asymmetric separation                                | Critical     | 6-DOF dynamic analysis + CG/inertia update + post-release control assessment                             |
| Release mechanism failure                            | Critical     | Independent path + defined safe fallback; no assumption of perfect release                               |
| Sensor loss or bad data                              | Hazardous    | Validity monitoring + redundant/diverse sensing + safe-state logic                                       |
| L1 automatic detection failure                       | Hazardous    | Independent L2 backup + L3 manual crew initiation                                                        |
| Power-bus failure                                    | Critical     | Independent survival power for critical functions                                                        |
| Communication loss                                   | Hazardous    | Local deterministic safety logic; communication not in authorization path                                |
| AI/ML model failure                                  | Hazardous    | Deterministic Safety Kernel remains authoritative                                                        |
| Fire during separation                               | Critical     | Fuel isolation + fire containment + coordinated sequencing                                               |
| Released component collision with capsule/parachutes | Catastrophic | Trajectory analysis + clearance envelopes + controlled sequencing                                        |
| Parachute deployment outside envelope                | Critical     | Hard inhibits + validated deployment conditions                                                          |
| Landing-support failure (one or more)                | Hazardous    | 8-support redundancy + asymmetric/failed-leg structural analysis                                         |
| Airbag inadvertent deployment in flight              | Critical     | Deterministic inhibit logic + dual-path sensing + mechanical safing                                      |
| Airbag non-deployment on impact                      | Hazardous    | Backup initiation path + sled test verification                                                          |
| Airbag overpressure / gas toxicity in sealed capsule | Critical     | OI-18: sealed-volume gas analysis required                                                               |
| Seat stroke bottom-out at extreme load               | Hazardous    | Progressive foam/absorber + support stroke + analysis against worst-case velocity                        |
| Common-cause EMI/lightning                           | Critical     | CCA + physical/electrical segregation + zone separation                                                  |
| Common-cause fire                                    | Critical     | Fire zone segregation + barriers between survival-critical systems                                       |

## 16. Principal System Interfaces

| ID        | Definition                                       | Normal-State                                         | Emergency Function                                    | ICD     |
|-----------|--------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|---------|
| CAP-AF    | Capsule ↔ airframe structural attachment         | Transmit normal flight loads                         | Controlled release on Safety Kernel authorization     | Doc 136 |
| CAP-PAR   | Capsule ↔ parachute system                       | Parachutes stowed; deployment inhibited              | Deploy primary canopies after clearance verified      | Doc 164 |
| CAP-LG    | Capsule ↔ eight landing supports                 | Ground loads during normal operations                | Primary crash/landing load path after separation      | Doc 156 |
| CAP-AB    | Capsule ↔ 16 airbags + sensors/inflators         | Stowed; armed only under deterministic preconditions | Deploy under Safety Kernel authorization at impact    | Doc 145 |
| CAP-SEAT  | Capsule floor ↔ 8 crashworthy seats + restraints | Normal seating load path                             | Stroke + energy absorption during impact              | Doc 123 |
| JET-PWR   | Independent release power bus                    | Standby / trickle charge                             | Power to all release mechanisms on command            | Doc 407 |
| JET-CMD   | Automatic/backup/manual command paths            | Inhibited; monitoring only                           | Transmit release commands through independent paths   | Doc 406 |
| JET-SNS   | Release-state sensing                            | Report attached state                                | Confirm released state at each stage                  | Doc 408 |
| FUEL-IS O | Fuel isolation ↔ separation coordination         | Normal fuel flow                                     | Isolate fuel before or concurrent with separation     | Doc 104 |
| REC-LOC   | Post-landing locator/communications              | Dormant / periodic beacon check                      | Activate post-separation; transmit position           | Doc 405 |
| AV-CTRL   | Avionics/control ↔ Safety Kernel                 | Pass flight data for monitoring                      | Safety Kernel issues authorization; avionics executes | Doc 241 |

## PART III

# Intellectual Property and Competitive Position

Candidate IP claims, prior-art posture, IP-to-engineering traceability, and competitive differentiation

## 17. Engineering / IP Value Surfaces

Rev 20/22/23 concentrates on four primary claim candidates and retains remaining elements as supporting system narrative. Four narrowly-scoped, well-supported claims are stronger than nine broad assertions.

### IP-A — Deterministic Safety Kernel Gating AI-Assisted Emergency Prediction

The specific combination of: (a) AI/ML-based anomaly prediction; (b) a hard deterministic precondition layer; and (c) selective, state-dependent (not all-or-nothing) actuation in a crewed rotorcraft survival system. Assessed as the strongest single claim in the package.

### IP-B — Eight-Point Fixed Capsule-Integrated Landing Support with Asymmetric Failure State Management

The combination of: (a) eight-point, capsule-integrated, fixed, energy-absorbing mechanical support configuration; (b) individual support failure-state monitoring; and (c) state management adapting stability assessment to asymmetric failure configurations.

### IP-C — State-Dependent Non-Rotor Selective Separation Sequencing for an Occupied Survival Capsule

Staged, state-dependent sequencing logic for selective release of non-rotor hazards from an occupied survival capsule, with explicit exclusion of the main rotor assembly and capsule-centric clearance verification gating recovery deployment. Distinguished from Ka-50/Ka-52 prior art.

### IP-D — Integrated Multi-Layer Survival Chain for Crewed Rotorcraft

The specific combination of: (a) Deterministic Safety Kernel (IP-A); (b) state-dependent non-rotor selective separation (IP-C); (c) eight-point capsule-integrated impact attenuation with asymmetric failure management (IP-B); and (d) distributed canopy recovery — as a unified, interdependent system architecture applied to a crewed rotorcraft survival capsule. System-level claims are harder to design around. Prior art (Ka-50: rotor separation + ejection seat; Soyuz: capsule + canopies) cannot anticipate the specific four-element combination. Assessed as the most commercially valuable claim in the package.

■ *Supporting narrative (not standalone patent candidates): six-layer taxonomy; multi-mode recovery; partial/asymmetric-release management; immutable emergency event record; 16-airbag layout; crashworthy seat stroke specification. No novelty, patentability or freedom-to-operate conclusion is asserted by this baseline.*

## 18. Risk Register

| Risk                                                                                | Severity | Current Status                | Priority           |
|-------------------------------------------------------------------------------------|----------|-------------------------------|--------------------|
| Stage S3 (capsule-to-airframe release) has no direct certified engineering analog   | High     | Open research item            | <b>P0 Critical</b> |
| Low-altitude canopy deployment envelope may exclude most real accident profiles     | High     | Open — closure study required | <b>P0 Critical</b> |
| 7,000 kg capsule mass not yet bottom-up verified                                    | Medium   | Planning allocation only      | <b>P0 Critical</b> |
| No 6-DOF simulation evidence for any separation stage                               | High     | Gate G4 not started           | <b>P0 Critical</b> |
| Capsule clearance from rotor disk after separation not analyzed                     | High     | Not yet addressed             | <b>P0 Critical</b> |
| 16-airbag deployment in sealed capsule — overpressure and gas-toxicity not analyzed | Critical | OI-18                         | <b>P0 Critical</b> |
| Airbag system mass 200 kg planning — detailed design may exceed budget              | Medium   | Corrected estimate            | <b>P0 High</b>     |
| Crashworthy seat design — stroke, load-distribution, combined-load cases not closed | Medium   | <b>TARGET/STUDY</b>           | <b>P0 High</b>     |
| Six-layer independence against common-cause not demonstrated                        | High     | CCA not performed             | <b>P0 Critical</b> |
| No professional prior-art search for IP-A, IP-B, IP-C, IP-D                         | Medium   | Prior-art posture only        | <b>P1 High</b>     |
| Certification basis not formally engaged with any regulator                         | Medium   | Conceptual reference          | <b>P1 High</b>     |
| Main-rotor separation (deferred) remains structurally unanalyzed                    | Medium   | Deferred research track       | <b>P2 Medium</b>   |

## 19. Prior Art Posture

### Class 1 — Ejection-seat rotorcraft (Ka-50/Ka-52)

Rotor-blade-severing + crew ejection is directly relevant prior art for any main-rotor-related separation claim. This is why main-rotor separation is excluded from all primary IP claims (IP-A through IP-D).

### Class 2 — Crewed spacecraft return capsules

Multi-canopy recovery (Soyuz 3 mains, Crew Dragon 3 mains), crashworthy seating, and independent post-landing life support are prior art for capsule-recovery and life-support elements. IP claims are narrowed to the rotorcraft-specific combination, not the general capsule concept.

### Class 3 — Helicopter energy-absorbing landing gear

Energy-absorbing skids and struts exist in certified rotorcraft. IP-B novelty rests on the eight-point capsule-integrated configuration and individual support failure-state management — not on energy absorption in general.

### Class 4 — Autonomous vehicle safety interlocks (IEC 61508, ISO 26262, DO-178C)

Deterministic safety interlocks are well developed. IP-A novelty is not in the concept of a safety interlock but in the specific combination of AI prediction + deterministic kernel + selective capsule-centric actuation in a crewed rotorcraft survival system.

Class 5 — Automotive and aviation airbag systems

Airbag systems are prior art. Any IP related to airbags would rest on the capsule-specific integration with the Safety Kernel, the sealed-volume environment, and the combined-attenuation architecture.

Class 6 — IP-D system-level combination

No single prior-art reference combines all four elements of IP-D (Safety Kernel + selective non-rotor separation + eight-point capsule attenuation + distributed canopy recovery) simultaneously in a crewed rotorcraft architecture. This is the basis for IP-D's strength. A professional prior-art search against IP-A through IP-D is required before any patent filing or broad non-NDA disclosure.

23. Competitive Differentiation

■ Descriptive comparison, not a claim of superior certified safety performance. All SHS advantages are architectural concepts pending engineering verification.

| Feature                                              | Bell 412 | AW139 | S-92 | Mi-26    | SHS Rev23      |
|------------------------------------------------------|----------|-------|------|----------|----------------|
| Occupied survival capsule as final landing boundary  | X        | X     | X    | X        | ✓ CONCEPT      |
| Deterministic Safety Kernel (AI-gated)               | X        | X     | X    | X        | ✓ CONCEPT      |
| State-dependent selective non-rotor separation       | X        | X     | X    | X        | ✓ CONCEPT      |
| Eight-point capsule-integrated impact attenuation    | X        | X     | X    | X        | ✓ CONCEPT      |
| Distributed canopy recovery (3+2)                    | X        | X     | X    | X        | ✓ STUDY        |
| 16-airbag occupant protection in capsule             | X        | X     | X    | X        | ✓ TARGET/STUDY |
| Six independently-capable survival layers            | X        | X     | X    | X        | ✓ CONCEPT      |
| Independent post-separation life support             | X        | X     | X    | X        | ✓ TARGET       |
| Explicit asymmetric/partial-failure state management | X        | X     | X    | X        | ✓ CONCEPT      |
| Crashworthy seats (≥150 mm stroke)                   | Some     | Some  | Some | Military | ✓ TARGET/STUDY |

**Commercial positioning.** SHS is not positioned as a replacement for existing helicopter models on cabin-space or payload-fraction grounds. It is positioned as a distinct product category — a survivability-first rotorcraft — for operators whose primary requirement is occupant survival in catastrophic emergencies: offshore energy, search-and-rescue, medical evacuation, governmental/military VIP transport.

## PART IV

# Critical Open Items Register

The complete OI-01 through OI-18 register with closure paths, and the IP-to-engineering traceability matrix

## 20. Critical Open Items Register

| ID    | Open Item                                                                           | Status          | Pri | Closure Path                                            |
|-------|-------------------------------------------------------------------------------------|-----------------|-----|---------------------------------------------------------|
| OI-01 | Stage S3 capsule-to-airframe release has no certified engineering analog            | OPEN — CRITICAL | P0  | Mechanism design + 6-DOF + ground demo                  |
| OI-02 | Minimum reliable canopy deployment altitude for 7,000 kg capsule not closed         | OPEN — CRITICAL | P0  | Sizing + inflation dynamics + envelope study            |
| OI-03 | 7,000 kg capsule mass remains a planning allocation                                 | OPEN — CRITICAL | P0  | Component mass database + weighing plan                 |
| OI-04 | No 6-DOF simulation for any separation stage                                        | OPEN — CRITICAL | P0  | Model hierarchy + correlation plan (G4)                 |
| OI-05 | Partial and asymmetric separation detection and degraded recovery not detailed      | OPEN — CRITICAL | P0  | State sensing + FMEA + test cases                       |
| OI-06 | Independence of six safety layers against common-cause not demonstrated             | OPEN — CRITICAL | P0  | CCA + fault injection + power/sensor diversity          |
| OI-07 | Capsule clearance from rotor disk after separation not analyzed                     | OPEN — CRITICAL | P0  | 6-DOF clearance analysis                                |
| OI-08 | Occupant injury criteria and seat/restraint dynamic performance not closed          | OPEN            | P0  | Dynamic seat tests + criteria selection                 |
| OI-09 | Eight-support asymmetric failure and terrain interaction cases not closed           | OPEN            | P0  | FEA + drop test matrix                                  |
| OI-10 | 26,000 kg vehicle mass lacks component-level uncertainty tracking                   | OPEN            | P1  | Mass control plan + reserve policy                      |
| OI-11 | Rotor/propulsion power-available vs power-required (OEI) not closed                 | OPEN            | P1  | Performance model + engine data                         |
| OI-12 | Fuel isolation timing coordinated with separation not closed                        | OPEN            | P1  | Hazard analysis + sequence timing                       |
| OI-13 | Professional prior-art search for IP-A through IP-D not performed                   | OPEN            | P1  | External IP counsel                                     |
| OI-14 | Certification basis not formally engaged                                            | OPEN            | P1  | Regulator engagement plan                               |
| OI-15 | ICDs for CAP-AF, CAP-PAR, CAP-LG, CAP-AB, CAP-SEAT, JET-* not issued                | OPEN            | P1  | ICD set under configuration control                     |
| OI-16 | 16-airbag packaging, placement, mass, inflators, deployment logic not closed        | OPEN            | P0  | Design + sled/impact tests                              |
| OI-17 | Crashworthy seat stroke and load design ( $\geq 150$ mm, $\leq 6,670$ N) not closed | OPEN            | P0  | Seat design + dynamic tests                             |
| OI-18 | 16-airbag deployment in sealed capsule — overpressure and gas-toxicity not analyzed | OPEN — CRITICAL | P0  | Sealed-volume gas analysis + toxicity + venting/staging |

### 33. IP-to-Engineering Traceability Matrix

This matrix connects the four principal IP surfaces to the engineering domains that must close before any claim can be treated as more than conceptual.

| IP ID | Claim Surface                                                                                                                 | Engineering Domains That Must Close                                                                                             | Primary Tree Anchors             | Critical Dependencies                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------|
| IP-A  | Deterministic Safety<br>Kernel gating<br>AI-assisted emergency prediction                                                     | Safety Kernel requirements;<br>precondition set;<br>independence from AI/ML;<br>fault injection;<br>software/hardware assurance | SAF / AV / VER series            | OI independence + formal verification |
| IP-B  | Eight-point fixed capsule-integrated landing support with asymmetric failure state management                                 | Support geometry; energy absorbers; failed-leg FEA; drop tests; terrain cases                                                   | Landing-support / crash / STRUCT | OI-09 + G5 evidence                   |
| IP-C  | State-dependent non-rotor selective separation sequencing for occupied survival capsule                                       | Candidate component list; S0–S5; release mechanisms; clearance verification; 6-DOF                                              | SEP / DYN series                 | OI-01, OI-04, OI-05, OI-07            |
| IP-D  | Integrated multi-layer survival chain (Kernel + selective separation + eight-point attenuation + distributed canopy recovery) | All of IP-A/B/C evidence plus recovery envelope and combined system demonstration                                               | Cross-domain G3–G6               | System-level integration gates        |

### 34. Buyer Due-Diligence Readiness Matrix

Intended to let an industrial acquirer review the baseline without reconstructing the entire programme architecture from the full document tree.

| Due-Diligence Question                 | HSH-00 Answer                                                                                                                                                                      | Primary Supporting Documents                     | Closure State           |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------|
| What is the core system proposition?   | Six-layer survival architecture centered on an occupied survival capsule, deterministic Safety Kernel, selective non-rotor separation, 8-point impact attenuation and 3+2 recovery | HSH-00 Part I §2, Part II §6, §12; Part VIII §30 | Defined                 |
| What is technically defined today?     | Top-level geometry, mass allocations, occupancy lock, Safety Kernel preconditions, S0–S5 concept, airbag/seat architecture, IP-A–D posture, OI-01–OI-18                            | HSH-00 Parts I–IX                                | Defined / Studied       |
| What is not yet proven?                | The open-item register remains the evidence gap, especially S3 release, low-altitude recovery, mass bottom-up, 6-DOF, OI-18 gas analysis                                           | HSH-00 Part III §18, Part IV §20                 | Open                    |
| What is the evidence path?             | G0 architecture freeze through G8 certification/qualification; E0–E5 evidence ladder                                                                                               | HSH-00 Part VII §26, §37; VER series             | Defined / Not started   |
| What are the principal residual risks? | S3 release, low-altitude recovery envelope, 7,000 kg mass closure, sealed-capsule airbag environment, six-layer independence                                                       | HSH-00 Part III §18, Part IV §20                 | Open / Critical         |
| What is the IP posture?                | IP-A through IP-D are candidate conceptual surfaces with prior-art awareness; not filed; professional search required                                                              | HSH-00 Part III §17, §19                         | Conceptual / Legal open |

| Due-Diligence Question        | HSH-00 Answer                                                                             | Primary Supporting Documents        | Closure State |
|-------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------|---------------|
| What configuration is locked? | Generation 1 maximum occupancy of 8 persons; no main-rotor separation in core safety case | HSH-00 Part I §1, §2; Part VIII §28 | Controlled    |

PART V

Performance, Technology Readiness and Mass Accounting

Hover-power and mission analysis, TRL assessment, mass/CG breakdown, and preliminary capsule geometry

21. Preliminary Performance Analysis

21.1 Hover Power Study Calculation

Using actuator disk theory:  $P_{ideal} = T^{3/2} / \sqrt{(2\rho A)}$ , where  $T = 26,000 \times 9.81 = 255,060 \text{ N}$ ;  $\rho = 1.225 \text{ kg/m}^3$  (ISA sea level);  $A = 530.93 \text{ m}^2$ .  $P_{ideal} \approx 3.57 \text{ MW}$  [CALC] — a study value; figure of merit  $FM = 0.75$  gives  $P_{shaft} \approx 4.76 \text{ MW}$  [CALC]. Installed engine power must exceed  $P_{shaft}$  to account for profile power, transmission losses (~4–8%), tail-rotor power (~8–12% of main rotor), accessory extraction, OEI margins, and hot/high performance. A plausible total installed power requirement is in the range of 6–8 MW, corresponding to two turboshaft engines of approximately 3,000–4,000 kW each — consistent with this mass class.

21.2 Mission Profile

| Mission Parameter     | Target Value          | Status   | Note                                      |
|-----------------------|-----------------------|----------|-------------------------------------------|
| Design range          | 1,000 km              | TARGET   | Must be closed by mission fuel model      |
| Crew + occupants      | 2 crew + 8 passengers | TARGET   | 8-person capsule limit                    |
| Cruise speed          | TBD                   | OPEN     | Depends on rotor/drag/engine selection    |
| Cruise altitude       | TBD                   | OPEN     | Optimized for range/power balance         |
| Takeoff fuel          | TBD                   | OPEN     | Included in 3,400 kg fuel budget          |
| Cruise fuel           | TBD                   | OPEN     | Dominant range-driver                     |
| Reserve policy        | TBD                   | OPEN     | Per applicable civil/military requirement |
| IFR diversion reserve | TBD                   | OPEN     | Typically 20–30 min additional fuel       |
| Unusable fuel         | ~120 kg               | PLANNING | Included in fuel budget                   |
| OEI diversion case    | Required              | REQ      | Safety-case requirement                   |
| Hot/high performance  | Required              | REQ      | ISA +20°C at 3,000 m minimum target       |

■ The 1,000 km range at 26,000 kg MTOW with two turboshaft-class engines is achievable within the 3,400 kg fuel allocation if cruise SFC is in the range of 0.25–0.35 kg/(kW·h) — consistent with modern turboshaft-class engines — but requires closure of aerodynamic drag, rotor performance, and engine selection. No range claim is made until this closure is complete.

22. Technology Readiness Levels

TRL assessment per the NASA/ESA TRL scale (1–9). Values are engineering judgments pending formal TRL review.

| System / Subsystem                      | TRL Today | Basis for Assessment              | Target G2 | Target G8 |
|-----------------------------------------|-----------|-----------------------------------|-----------|-----------|
| Six-layer survival architecture concept | TRL 2     | Architecture defined; no hardware | TRL 4     | TRL 8     |

| System / Subsystem                           | TRL Today | Basis for Assessment                                      | Target G2          | Target G8 |
|----------------------------------------------|-----------|-----------------------------------------------------------|--------------------|-----------|
| Deterministic Safety Kernel (IP-A)           | TRL 2     | Precondition set defined; no software                     | TRL 4              | TRL 8     |
| Capsule CFRP roll-cage concept               | TRL 2–3   | Similar CFRP capsule structures exist in spacecraft       | TRL 5              | TRL 8     |
| Eight fixed landing supports                 | TRL 2     | Similar energy-absorbers exist; capsule-integration novel | TRL 4              | TRL 7     |
| Stage S0–S2 (arming, fuel isolation)         | TRL 3     | Analogous systems exist in military jettison hardware     | TRL 5              | TRL 8     |
| Stage S3 (capsule-to-airframe release)       | TRL 1     | No direct analog identified                               | TRL 3              | TRL 7     |
| 3+2 canopy recovery                          | TRL 3     | Three-main canopy architecture proven in spacecraft       | TRL 5              | TRL 7     |
| Crashworthy seats ( $\geq 150$ mm stroke)    | TRL 3–4   | Analogous seats exist in military helicopters             | TRL 5              | TRL 7     |
| 16-airbag capsule architecture               | TRL 1–2   | Airbag technology mature; capsule-integration novel       | TRL 3              | TRL 6     |
| Sealed-capsule airbag gas management (OI-18) | TRL 1     | No analogous system identified                            | TRL 3              | TRL 6     |
| Mass/CG/inertia model                        | TRL 2     | Planning model only                                       | TRL 5              | TRL 8     |
| 6-DOF separation dynamics model              | TRL 1     | No simulation started                                     | TRL 4              | TRL 8     |
| IP-A through IP-D patent claims              | TRL 1     | Architecture defined; no legal filing                     | Prior art searched | Filed     |

## PART VI

# Geometry and Mass Accounting

Top-level mass tables, capsule component mass, detailed aircraft mass breakdown, mass-closure confidence, and preliminary capsule geometry

## Top-Level Technical Baseline — Master Table

| Parameter                 | Rev 23 Value                 | Status       | Key Note                            |
|---------------------------|------------------------------|--------------|-------------------------------------|
| Design mass               | 26,000 kg                    | TARGET       | Planning closure; bottom-up pending |
| Maximum capsule occupancy | 8 persons                    | LOCKED LIMIT | Generation 1 hard limit             |
| Main rotor diameter       | 26.0 m                       | TARGET       | Benchmark comparison §4.1           |
| Main rotor disk area      | 530.93 m <sup>2</sup>        | CALC         | $\pi \times (13.0)^2$               |
| Disk loading              | 48.97 kg/m <sup>2</sup>      | CALC         | See §4.1                            |
| Tail rotor diameter       | 4.0 m                        | TARGET       | Anti-torque closure pending         |
| Tail rotor disk area      | 12.57 m <sup>2</sup>         | CALC         | $\pi \times (2.0)^2$                |
| Capsule mass              | 7,000 kg                     | TARGET       | Bottom-up pending Doc 113           |
| Range target              | 1,000 km                     | TARGET       | Mission model pending §21           |
| Ideal hover power         | ≈ 3.57 MW                    | CALC         | Actuator disk, ISA SL               |
| Shaft power @ FM 0.75     | ≈ 4.76 MW                    | CALC         | Study value only                    |
| Propulsion                | Two turboshaft engines       | REQ/STUDY    | Engine selection open               |
| Overall length            | 16.5–18.0 m                  | TARGET/STUDY | Layout reconciliation pending       |
| Fuselage width (max)      | 4.0–4.5 m                    | TARGET/STUDY |                                     |
| Overall height            | 4.8–5.3 m                    | TARGET/STUDY |                                     |
| Main rotor separation     | Not part of core safety case | DEFERRED     | §12                                 |
| Recovery canopies         | 3 primary + 2 reserve        | STUDY        | §11                                 |
| Separation timeline       | Staged S0–S5                 | TARGET/OPEN  | §13                                 |
| Occupant airbags          | 16 total (2 per occupant)    | TARGET/STUDY | §10A; OI-18                         |
| Crashworthy seat stroke   | ≥150 mm dynamic stroke       | TARGET/STUDY | §10B                                |
| Peak spinal load limit    | ≤6,670 N reference criterion | TARGET/STUDY | FAR 27.562 proxy; §10B              |
| Development gate          | G0 approaching closure       | CURRENT      | G1–G8 not started                   |

## Mass Accounting — 26,000 kg

| Mass Group                                        | Planning Mass    | Status                         |
|---------------------------------------------------|------------------|--------------------------------|
| Survival capsule, complete                        | 7,000 kg         | TARGET                         |
| Main airframe excluding capsule                   | 5,000 kg         | ASSUMPTION/STUDY               |
| Main rotor, hub and controls                      | 2,100 kg         | ASSUMPTION/STUDY               |
| Tail rotor, gearbox and drive                     | 480 kg           | ASSUMPTION/STUDY               |
| Engines, transmission, gearboxes and mounts       | 3,100 kg         | ASSUMPTION/STUDY               |
| Fuel system + full usable fuel                    | 3,400 kg         | ASSUMPTION/STUDY               |
| Rapid separation system                           | 1,000 kg         | ASSUMPTION/STUDY               |
| Avionics, electrical, hydraulic, thermal, control | 1,600 kg         | ASSUMPTION/STUDY               |
| Operating fluids and consumables                  | 320 kg           | ASSUMPTION/STUDY               |
| Growth / configuration reserve                    | 2,000 kg         | PROGRAM RESERVE                |
| <b>TOTAL</b>                                      | <b>26,000 kg</b> | <b>CALC/ACCOUNTING CLOSURE</b> |

## Capsule Component Mass — 7,000 kg

| Capsule Component                                                            | Planning Mass   | Notes                            |
|------------------------------------------------------------------------------|-----------------|----------------------------------|
| Primary crashworthy cell / internal roll-cage                                | 1,950 kg        | CFRP + titanium nodes            |
| Outer capsule shell / fairing / thermal protection                           | 500 kg          |                                  |
| Eight fixed landing supports + progressive energy absorbers                  | 850 kg          |                                  |
| Eight crashworthy seats + restraints ( $\geq 150$ mm stroke, $\leq 6,670$ N) | 700 kg          | Rev 20 corrected spec            |
| Sixteen occupant airbags + inflators + sensors + wiring                      | 200 kg          | Rev 20 corrected (was 80–120 kg) |
| Emergency exits / doors, frames and seals                                    | 250 kg          |                                  |
| Distributed multi-canopy recovery system (3 primary + 2 reserve)             | 650 kg          |                                  |
| Independent capsule emergency power + batteries                              | 250 kg          |                                  |
| Life support / emergency oxygen / environmental                              | 200 kg          |                                  |
| Avionics / sensing / communications / locator / lighting                     | 250 kg          |                                  |
| Separation interfaces + capsule-side release hardware                        | 350 kg          |                                  |
| Internal secondary structure / floor / partitions / padding                  | 300 kg          | Reduced to absorb airbag mass    |
| Fire / smoke detection and suppression                                       | 150 kg          |                                  |
| Water landing / flotation / sealing                                          | 150 kg          |                                  |
| Inspection / access / replaceable protection                                 | 100 kg          |                                  |
| Capsule configuration growth allowance                                       | 150 kg          |                                  |
| <b>TOTAL</b>                                                                 | <b>7,000 kg</b> | <b>CALC/CLOSURE</b>              |

■ Airbag system mass corrected from 80–120 kg to 200 kg based on automotive-class benchmarks scaled to a 16-bag aviation-grade system with ECU, pyrotechnic harness, mounting and qualification margin. Corresponding reduction taken from internal secondary structure. Total capsule budget unchanged at 7,000 kg. If detailed design exceeds 200 kg, growth uses the 150 kg allowance; further growth triggers configuration-controlled impact assessment.

## 24. Preliminary Capsule Geometry Estimate

Shows geometric plausibility of 8 occupants + airbags + crashworthy seats in a capsule of 7,000 kg. All values pending Document 112 closure.

| Geometry Parameter                             | Preliminary Estimate                                  | Status |
|------------------------------------------------|-------------------------------------------------------|--------|
| Capsule outer diameter                         | ≈ 3.2–3.6 m                                           | STUDY  |
| Capsule internal usable diameter               | ≈ 2.6–3.0 m (after roll-cage wall thickness)          | STUDY  |
| Capsule internal height (usable)               | ≈ 1.75–2.0 m (seated + headroom)                      | STUDY  |
| Capsule outer shell thickness (CFRP + fairing) | ≈ 100–150 mm per side                                 | STUDY  |
| Roll-cage member cross-section                 | ≈ 60–100 mm diameter CFRP tubes                       | STUDY  |
| Seat arrangement                               | 2 rows × 4 seats, or circular/staggered 8-seat layout | OPEN   |
| Seat pitch                                     | ≈ 650–750 mm                                          | STUDY  |
| Seat width                                     | ≈ 450–500 mm per position                             | STUDY  |
| Seat stroke height occupied by absorber        | ≈ 150–200 mm below seat pan                           | STUDY  |

A circular capsule of ~3.4 m outer diameter, ~2.8 m internal diameter, ~1.85 m internal height accommodates 8 crashworthy seats in a 2×4 configuration with ≈700 mm pitch; 16 airbags stowed in headrests, sill-mounted and seat-back positions; 5-point restraints; emergency exits at a minimum of 2 locations with airbag-compatible egress paths; and internal volume for life support, emergency power, avionics and fire suppression. Geometric plausibility is supported at this preliminary level — bottom-up closure is required (Documents 112, 113).

## 25. Detailed Aircraft Mass Breakdown

### Main Airframe — 5,000 kg

Fuselage primary load-bearing structure 1,450 kg; frames, bulkheads and longerons 650 kg; tail boom and tail structural assemblies 520 kg; engine and transmission support structure 380 kg; rotor mast/transmission support integration 260 kg; external skins, fairings and aerodynamic panels 310 kg; doors, access panels and hinges 190 kg; firewalls, barriers and structural fire protection 170 kg; structural fittings, lugs, fasteners and local reinforcements 260 kg; cabin/service floor structure outside capsule boundary 210 kg; integration margin remainder to 5,000 kg. All values **ASSUMPTION/STUDY** pending bottom-up closure.

### Main Rotor System — 2,100 kg

Main rotor blades (26.0 m system, 4–5 blades) 980 kg; rotor hub and blade retention/grips 280 kg; mast and rotor-head structural components 260 kg; swashplate and pitch-control hardware 170 kg; pitch links, control rods and bearings 110 kg; rotor vibration/dynamic-control hardware 80 kg; rotor lubrication/inspection/access 40 kg; configuration allowance + integration margin remainder. All values **ASSUMPTION/STUDY** pending bottom-up closure.

### Tail Rotor System — 480 kg

Tail rotor blades (4.0 m) 145 kg; hub and blade retention 65 kg; tail rotor gearbox 95 kg; drive shafting, couplings and bearings 60 kg; controls and pitch mechanism 35 kg; tail structure/interface hardware + margin remainder. All values **ASSUMPTION/STUDY** pending bottom-up closure.

## Engines and Transmission — 3,100 kg

Engine #1 complete installed 700 kg; Engine #2 complete installed 700 kg; engine accessories and engine-control hardware 180 kg; main gearbox / combining transmission 650 kg; intermediate/reduction transmission elements 180 kg; drive shafts and couplings 100 kg; lubrication, oil cooling and filtration 80 kg; engine/gearbox mounts and vibration isolation 60 kg; exhaust and hot-section installation 50 kg; propulsion integration margin remainder. All values **ASSUMPTION/STUDY** pending bottom-up closure.

## Fuel System + Usable Fuel — 3,400 kg

Usable fuel mass ~2,100 kg; fuel tanks (crash/fire-resistant structure) 280 kg; fuel pumps and boost/transfer units 90 kg; fuel lines, hoses, valves and fittings 110 kg; fuel filters, sensors and quantity measurement 55 kg; fuel isolation / emergency shutoff hardware 75 kg; venting, surge and inerting/fire-protection 70 kg; unusable / residual fuel allocation 120 kg; fuel system integration margin remainder. All values **ASSUMPTION/STUDY** pending bottom-up closure.

## Avionics / Electrical / Hydraulic / Thermal — 1,600 kg

Flight-control computers 180 kg; flight-control actuators/servos 210 kg; navigation/air-data/flight sensors 100 kg; communications/mission avionics 90 kg; electrical generation/distribution 190 kg; batteries/emergency electrical 90 kg; wiring/harnesses/connectors 170 kg; hydraulic pumps/reservoirs/lines/actuators 180 kg; thermal management/cooling 100 kg; environmental control/ventilation 70 kg; fire detection/warning/suppression interfaces 55 kg; health monitoring/data recording 45 kg; lighting/service/miscellaneous 20 kg; integration margin remainder. All values **ASSUMPTION/STUDY** pending bottom-up closure.

## 39. Mass Closure Confidence Matrix

Controlled 26,000 kg accounting model and 7,000 kg capsule allocation with a closure-confidence view so each major mass block has a defined route from allocation to evidence.

| Mass Block                                  | Rev 20/23 Planning | Current Basis       | Closure Evidence                                      | Tree Anchor       | Impact |
|---------------------------------------------|--------------------|---------------------|-------------------------------------------------------|-------------------|--------|
| Complete survival capsule                   | 7,000 kg           | Planning allocation | Bottom-up component database + weighed articles       | MASS-003; capsule | High   |
| Main airframe                               | 5,000 kg           | Assumption / study  | Bottom-up structural and systems mass model           | MASS-004          | High   |
| Main rotor / hub / controls                 | 2,100 kg           | Assumption / study  | Rotor component mass model + design definition        | MASS-005; ROT     | High   |
| Tail rotor / gearbox / drive                | 480 kg             | Assumption / study  | Component definition + supplier / design data         | ROT / PROP        | Medium |
| Engines / transmission / mounts             | 3,100 kg           | Assumption / study  | Selected engine and gearbox data + installation model | MASS-006; PROP    | High   |
| Fuel system + usable fuel                   | 3,400 kg           | Assumption / study  | Fuel accounting + tank/pump/line definition           | MASS-012; FUEL    | High   |
| Rapid separation system                     | 1,000 kg           | Assumption / study  | Mechanism and interface mass model                    | SEP domain        | High   |
| Avionics / electrical / hydraulic / thermal | 1,600 kg           | Assumption / study  | Subsystem mass models + harness estimate              | AV / ELEC         | Medium |
| Operating fluids / consumables              | 320 kg             | Assumption / study  | Fluid accounting                                      | MASS series       | Low    |
| Growth / configuration reserve              | 2,000 kg           | Program reserve     | Reserve policy + uncertainty tracking                 | Mass control plan | High   |
| Airbags (inside capsule 200 kg line)        | 200 kg             | Corrected estimate  | Module + ECU + harness + mounting bill of materials   | CAP-AB docs       | High   |

| Mass Block                     | Rev 20/23 Planning | Current Basis | Closure Evidence                           | Tree Anchor   | Impact |
|--------------------------------|--------------------|---------------|--------------------------------------------|---------------|--------|
| Crashworthy seats + restraints | 700 kg             | Planning line | Seat supplier data + stroke mechanism mass | CAP-SEAT docs | High   |

## PART VII

# Verification, Maturity and Evidence Architecture

Gate structure, evidence ladder, engineering maturity framework, engineering claim architecture, and subsystem TRL/evidence/gate matrix

## 26. Verification, Validation and Evidence Program

| Gate | Evidence Objective              | Key Evidence Items                                                                                                    | Status              |
|------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------|
| G0   | Architecture Freeze             | Safety philosophy; boundaries; interfaces; emergency states; airbag and seat concepts locked                          | Approaching closure |
| G1   | Analytical Closure              | Mass, CG, inertia, rotor, power, fuel, structural load paths, separation kinematics, seat-stroke analysis             | Not started         |
| G2   | Subsystem Demonstration         | Release mechanisms, locks, absorbers, parachute, airbag module, seat stroke, sensors, independent power               | Not started         |
| G3   | Integrated Ground Demonstration | End-to-end sequence; timing; state sensing; capsule clearance; airbag deployment in representative volume             | Not started         |
| G4   | 6-DOF Simulation                | Vehicle and capsule dynamics; partial/asymmetric release; parachute inflation; sealed-capsule airbag gas              | Not started         |
| G5   | Structural / Crash Testing      | Capsule, attachments, seats (dynamic stroke test per FAR 27.562 proxy), airbag sled tests, supports, energy absorbers | Not started         |
| G6   | Recovery System Testing         | Canopy deployment; inflation loads; failure cases; stability; landing performance                                     | Not started         |
| G7   | Flight Development              | Progressive envelope expansion; normal and emergency recovery validated in flight                                     | Not started         |
| G8   | Certification / Qualification   | Full evidence package against selected certification basis                                                            | Not started         |

■ **Terminology:** *Specified* = defined in design; *Analyzed* = documented analytical result; *Simulated* = supported by a model; *Tested* = demonstrated in a defined test; *Verified* = evidence shows compliance; *Validated* = suitability established for intended use; *Qualified/Certified* = formal processes complete. As of Rev 23: all HSH-00 claims are *Specified* or *Studied*. None are *Analyzed*, *Simulated*, *Tested*, *Verified*, *Validated*, *Qualified*, or *Certified*.

## 27. Engineering Maturity Framework

| Configuration Item      | Rev 23 Control Rule                                                  | Closure Evidence Required                          |
|-------------------------|----------------------------------------------------------------------|----------------------------------------------------|
| MTOW / design mass      | Freeze only after bottom-up mass model and reserve policy reconciled | Mass report + uncertainty analysis                 |
| Rotor geometry          | Maintain 26.0 m target until performance study closes                | Rotor sizing + power model                         |
| Capsule mass            | 7,000 kg planning target; no closure claim                           | Bottom-up mass budget (Doc 113)                    |
| Occupancy               | 8 persons maximum — LOCKED for Generation 1                          | Configuration control only                         |
| Recovery                | 3 primary + 2 reserve study architecture                             | Sizing + inflation dynamics + failure testing      |
| Separation              | Non-rotor candidate set only in core baseline                        | 6-DOF + clearance + ground demonstration           |
| Safety authorization    | Deterministic kernel remains authoritative                           | Requirements + software/hardware verification      |
| Airbags                 | 16 total (2 per occupant) TARGET/STUDY                               | Packaging + sled/impact tests + OI-18 gas analysis |
| Crashworthy seat stroke | ≥150 mm stroke; ≤6,670 N criterion                                   | Dynamic seat tests + analysis                      |

**Maturity roadmap:** Architecture control (G0) → Analytical closure (G1) → Subsystem design (G2 prep) → Ground demonstration (G3) → Model validation (G4) → Structural/recovery tests (G5/G6) → Flight development (G7) → Qualification/certification (G8).

## 32. Engineering Claim Architecture

Purpose: convert the principal HSH engineering and IP propositions into traceable engineering claims. Each claim is connected to the evidence that must exist before the claim can be treated as demonstrated. This is an architecture for closure, not a statement that the claims are already proven.

| ID             | Engineering Claim                                                                                                                                            | Primary Evidence Required                                                                         | Verification Path                                                                     | Basis               |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------|
| <b>EC-00 1</b> | Occupied survival capsule can become the final landing boundary in the defined emergency architecture                                                        | Requirements, structural load paths, attachment analysis, separation dynamics, clearance evidence | Analysis → 6-DOF simulation → ground demonstration → applicable testing               | Specified / Studied |
| <b>EC-00 2</b> | State-dependent selective separation can release designated non-rotor hazards while preserving capsule survivability                                         | Release mechanism design, authorization logic, partial/asymmetric cases, failed-release fallback  | Analysis → mechanism test → integrated ground demonstration                           | Specified / Open    |
| <b>EC-00 3</b> | Deterministic Safety Kernel can remain the authoritative authorization boundary for safety-critical actuation                                                | Requirements, precondition logic, independence analysis, formal/robustness evidence               | Requirements verification → formal methods / fault injection → qualification evidence | Specified / Open    |
| <b>EC-00 4</b> | Eight capsule-integrated landing supports can provide progressive impact attenuation with asymmetric failure management                                      | Load cases, FEA, absorber characteristics, terrain cases, failed-support redistribution           | Analysis → drop testing → integrated crashworthiness evidence                         | Specified / Open    |
| <b>EC-00 5</b> | 3 primary + 2 reserve canopy architecture can recover the 7,000 kg capsule within a validated deployment envelope                                            | Sizing, inflation dynamics, failure tolerance, minimum altitude envelope                          | Sizing analysis → deployment tests → envelope closure                                 | Study / Open        |
| <b>EC-00 6</b> | 16-airbag + ≥150 mm stroke seat architecture can limit occupant loads under defined impact conditions without defeating egress or sealed-capsule environment | Packaging, deployment logic, sled/impact tests, sealed-volume gas analysis (OI-18)                | Design → sled/impact → gas analysis → combined attenuation evidence                   | Target / Open       |

| ID            | Engineering Claim                                                                                     | Primary Evidence Required                                      | Verification Path                                             | Basis                  |
|---------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------|------------------------|
| <b>EC-007</b> | IP-A through IP-D remain coherent conceptual claim surfaces supported by the engineering architecture | Prior-art posture, claim definitions, engineering traceability | Professional prior-art search → claim drafting → legal review | Conceptual / Not filed |

## 37. Evidence Ladder

All HSH-00 claims are Specified or Studied. The ladder is the controlled progression from specification to demonstrated evidence.

| Ladder Step | Meaning                         | Minimum Evidence Content                                             | Typical Gate                  |
|-------------|---------------------------------|----------------------------------------------------------------------|-------------------------------|
| E0          | Specified / Studied             | Requirement or concept written with status label                     | HSH-00 (current)              |
| E1          | Analyzed                        | Documented analytical result with stated assumptions and uncertainty | G1 mass, loads, kinematics    |
| E2          | Simulated                       | Model result with stated fidelity and validation status              | G4 6-DOF; OI-18 gas model     |
| E3          | Subsystem tested                | Defined test article, conditions, instrumentation, results           | G2 mechanisms, seats, airbags |
| E4          | Integrated demonstrated         | End-to-end ground sequence under controlled conditions               | G3 integrated demonstration   |
| E5          | Flight / qualification evidence | Flight development and formal qualification against basis            | G7–G8                         |

## 38. Subsystem TRL + Evidence + Gate Matrix

Extends the TRL table by adding the evidence needed to move each subsystem toward programme gates. TRL values are not re-scored upward by documentation alone.

| Subsystem                              | Rev 20/23 TRL | Evidence Needed                                                | Next Gate | Target Evidence |
|----------------------------------------|---------------|----------------------------------------------------------------|-----------|-----------------|
| Six-layer survival architecture        | 2             | Independence analysis, CCA, fault injection                    | G1–G3     | E1–E4           |
| Deterministic Safety Kernel (IP-A)     | 2             | Requirements, formal/robust verification, independence from AI | G1–G2     | E1–E3           |
| Capsule CFRP roll-cage                 | 2–3           | FEA, joint/node design, crash load cases                       | G5        | E1–E3           |
| Eight fixed landing supports           | 2             | Absorber characterization, drop tests, asymmetric cases        | G5        | E2–E4           |
| Stage S0–S2 (arming, fuel isolation)   | 3             | Mechanism tests, timing, isolation verification                | G2–G3     | E3–E4           |
| Stage S3 (capsule-to-airframe release) | 1             | Mechanism design, 6-DOF, ground demonstration                  | G2–G4     | E1–E4           |
| 3+2 canopy recovery                    | 3             | Sizing, inflation, failure cases, min altitude                 | G6        | E2–E4           |
| Crashworthy seats ≥150 mm / ≤6,670 N   | 3–4           | Dynamic seat tests (FAR 27.562 proxy)                          | G2–G5     | E3–E4           |
| 16-airbag capsule architecture         | 1–2           | Packaging, deployment, sealed-volume gas, sled tests           | G2–G5     | E2–E5           |
| Mass / CG / inertia model              | 2             | Bottom-up mass database, uncertainty, CG/inertia closure       | G1        | E2              |

| Subsystem                 | Rev<br>20/23<br>TRL | Evidence Needed                                                          | Next Gate            | Target E<br>vidence |
|---------------------------|---------------------|--------------------------------------------------------------------------|----------------------|---------------------|
| 6-DOF separation dynamics | 1                   | Model specification, correlation plan                                    | G4                   | E3                  |
| IP-A through IP-D         | 1                   | Prior-art search, invention disclosures,<br>claim drafting, legal review | External IP<br>gates | E0–E1<br>legal      |

PART VIII

Master Configuration, Philosophy and Engineering Basis

Master configuration record, design philosophy, predecessor documents, external references, and the final Generation 1 baseline statement

28. Master Configuration Record

| Configuration Item        | Rev 23 Baseline                      | Status               |
|---------------------------|--------------------------------------|----------------------|
| Aircraft programme name   | SHS Heavy Safe Helicopter            | LOCKED               |
| Founder                   | Amirbahador Atai                     | CONTROLLED           |
| Generation                | Generation 1                         | LOCKED               |
| Document number           | SHS-00                               | LOCKED               |
| Revision                  | Rev 23 — Professional Master Edition | BASELINE             |
| Capsule occupancy maximum | 8 persons                            | LOCKED LIMIT         |
| Vehicle design mass       | 26,000 kg                            | TARGET/OPEN          |
| Capsule mass allocation   | 7,000 kg                             | PLANNING/OPEN        |
| Main rotor target         | 26.0 m                               | TARGET/OPEN          |
| Tail rotor target         | 4.0 m                                | TARGET/OPEN          |
| Range target              | 1,000 km                             | TARGET/OPEN          |
| Landing supports          | 8 fixed energy-absorbing             | CONCEPT/OPEN         |
| Recovery                  | 3 primary + 2 reserve canopies       | STUDY/OPEN           |
| Occupant airbags          | 16 (2 per occupant)                  | TARGET/STUDY         |
| Crashworthy seat stroke   | ≥150 mm; ≤6,670 N                    | TARGET/STUDY         |
| Core separation           | Selective, staged, non-rotor only    | CONCEPT/OPEN         |
| Main-rotor separation     | Excluded from core safety case       | DEFERRED RESEARCH    |
| Safety authorization      | Deterministic Safety Kernel          | CONCEPT/OPEN         |
| IP claims                 | IP-A, IP-B, IP-C, IP-D               | CONCEPTUAL/NOT FILED |
| Development gate          | G0 approaching closure               | CURRENT              |

40. Configuration Baseline Diagram (Textual)

Textual configuration map of the locked and target elements. Visual CAD is out of scope for this document.

| Configuration Class | Content                                                                             | Control Rule                        |
|---------------------|-------------------------------------------------------------------------------------|-------------------------------------|
| LOCKED LIMIT        | 8 persons maximum occupancy                                                         | Any change → full re-analysis       |
| LOCKED CONCEPT      | No retractable landing gear; capsule central; main rotor not in core separation set | Impact assessment required to alter |
| TARGET geometry     | 26.0 m main rotor; 4.0 m tail rotor; 26,000 kg design mass; 7,000 kg capsule        | Bottom-up and performance closure   |

| Configuration Class                          | Content                                                                                                                          | Control Rule                        |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| TARGET survival hardware                     | 8 fixed energy-absorbing supports; 16 airbags; ≥150 mm stroke seats ≤6,670 N; 3+2 canopies                                       | Design + test evidence              |
| NORMAL FLIGHT LOAD PATHS                     | Airframe ↔ CAP-AF ↔ capsule structure ↔ seats                                                                                    | Primary flight loads                |
| EMERGENCY LOAD PATH / FINAL LANDING BOUNDARY | Released capsule ↔ 8 supports ↔ crashworthy seats ↔ airbags ↔ occupants                                                          | Post-separation impact              |
| CONTROLLED INTERFACES                        | CAP-AF, CAP-PAR, CAP-LG, CAP-AB, CAP-SEAT, JET-*, FUEL-ISO                                                                       | ICD set under configuration control |
| STATE-DEPENDENT TRANSITION                   | Normal → Anomaly → Confirm → Assess → Recover if possible → Authorize → Selective separation → Clearance → Recovery → Landing    | Emergency state machine             |
| INDEPENDENCE PRINCIPLE                       | Physical, electrical and software segregation where required; human authority preserved; AI advisory only at the safety boundary | Non-negotiable                      |

■ Configuration-control anchor: any change to occupancy, mass, rotor geometry, separation architecture, recovery architecture, landing-support count/geometry, airbag count/placement, seat stroke or Safety Kernel preconditions triggers the impact-assessment process already defined in this baseline.

## 41. Document Dependency / Parent–Child Architecture

The Master Engineering Document Tree contains requirements, subsystem design, safety, dynamics, verification, certification, IP and programme-management documents. This section adds a dependency discipline so that each lower-level document can identify its controlling parent, upstream inputs, downstream verification artifact and closure gate.

| Document Layer         | Example Parent            | Example Child / Output                                           | Dependency Rule                                                                        |
|------------------------|---------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| HSH-00 Master Baseline | HSH-00 Rev 23             | REQ / SAF / MASS / ROT / SEP / REC / AV / DYN / VER / IP domains | Master baseline controls architecture, terminology, locked limits and gate definitions |
| System Requirements    | REQ-001–040               | Subsystem requirements, derived requirements, RVM                | Child requirements shall trace upward to parent REQ and to HSH-00                      |
| Safety domain          | SAF plans + FHA/PSSA/SSA  | CCA, PRA, ZSA, Safety Kernel verification                        | No safety claim without upward trace to HSH-00 layers and OI register                  |
| Capsule domain         | CAP requirements + design | Roll-cage, seats, airbags, supports, mass, tests                 | Must respect 8-person lock, 7,000 kg allocation, OI-16/17/18                           |
| Separation domain      | SEP requirements + design | Mechanisms, S0–S5, 6-DOF, ground demo                            | Main rotor remains excluded from core candidate set                                    |
| Recovery domain        | REC / PAR                 | 3+2 sizing, deployment, envelope                                 | OI-02 minimum altitude is critical path                                                |
| Verification domain    | VER plans + matrices      | G0–G8 evidence packages                                          | Evidence ladder E0–E5 applies                                                          |
| IP domain              | IP-A/B/C/D descriptions   | Prior-art search, claim drafts, FTO                              | No novelty assertion without external legal work                                       |

## 29. Engineering Basis and External References

**Predecessor documents.** LSH-00 Rev 3.2 (six-layer philosophy; independent initiation; controlled separation; eight supports; capsule-as-final-boundary concept); USCC Doc 00 Rev 1.3 (jettison sequence; deterministic safety authorization); HSC-00/HSH-00 Rev 12–22 (progressive engineering development and correction).

**External engineering references.** EASA CS-29 Amendment 12: large-rotorcraft certification standard; jettisonable configurations require clear trajectory, no damaging interaction, acceptable post-jettison stability and performance. FAR 27.562 / CS-27 dynamic seat-test requirements: proxy reference criterion for  $\leq 6,670$  N peak spinal compressive load used in the crashworthy-seat specification (§10B). MIL-STD-1290A: helicopter crashworthiness design criteria; relevant to capsule structural design, seat design, landing-support design and occupant injury criteria. NASA TRL scale: used for TRL assessment in §22.

## 30. Design Philosophy — Controlled Survival, Not a Single Device

HSH Generation 1 integrates multiple survival mechanisms into one state-aware architecture. The vehicle does not depend solely on a parachute, solely on autorotation, solely on a detachable cabin, or solely on automatic software. Removing main-rotor separation from the core safety case — introduced in Rev 13 and retained through Rev 23 — strengthens rather than weakens the layered proposition: a safety case that requires its riskiest, least-proven capability to succeed is not actually layered. Rev 20/22/23 extends the impact-attenuation layer with 16 airbags and crashworthy seats specified to the correct engineering standard — these are complementary protections, not substitutes for structural crashworthiness, landing supports, or recovery canopies. The strongest future version of HSH Generation 1 is the version in which every important claim has a visible path from requirement to model to test to controlled evidence. Documentation is not a substitute for evidence. The path from this baseline to G8 is long and requires real engineering work — but the architecture is defined clearly enough, honestly enough, and with sufficient IP specificity to serve as a credible starting point for that work.

## 31. Final Generation 1 Baseline Statement

HSH-00 Rev 23 is the definitive Generation 1 conceptual master baseline for the HSH Heavy Safe Helicopter. It is the document a technical due-diligence team from a potential IP licensee should receive as the starting point for evaluation.

**What is defined:** the full six-layer survival architecture; the Safety Kernel and its precondition set; the staged S0–S5 separation timeline with an honest confidence-basis assessment; the 3+2 canopy recovery architecture with preliminary sizing; the 8-point capsule-integrated landing-support system; the 16-airbag and crashworthy-seat occupant-protection architecture (with corrected engineering specifications); four IP claims (IP-A through IP-D) with prior-art awareness; a full open-items register (OI-01 through OI-18); a TRL assessment; a competitive-differentiation table; a preliminary capsule geometry check; a mission-profile framework; and the diligence enhancement layer of Part IX.

**What remains to be proven:** everything in the open-items register. The gap between architecture and evidence is the scope of the Generation 1 engineering document tree (Documents 001–500).

## PART IX

# Safety-Case Synthesis and Failure Containment

Layered safety-case argument, failure-containment philosophy, and buyer-facing due-diligence synthesis

## 35. Safety Case Architecture

The safety case is organized as a layered argument: the vehicle shall provide multiple independent means of protecting occupants, ordered from conventional recovery to capsule separation and post-landing survival. The safety case does not rely on main-rotor separation. Each layer requires defined requirements, hazards, verification methods and residual-risk acceptance.

| Safety-Case Block               | Argument                                                                | Evidence Required                                              | Principal Residual Risk                            |
|---------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------|
| Layer L1–L3 initiation          | Automatic primary, automatic backup, and non-bypassable crew authority  | Independence analysis, fault injection, HMI verification       | Common-cause defeat of all initiation paths        |
| Layer L4 selective separation   | State-dependent release of designated non-rotor hazards only            | Mechanism design, S0–S5 timing, 6-DOF, ground demonstration    | Unintended release; incomplete /asymmetric release |
| Layer L5 direct-impact survival | Capsule structure, roll-cage, 8 supports, crashworthy seats, 16 airbags | FEA, drop/sled tests, OI-18 gas analysis, injury criteria      | Bottom-out; overpressure; egress blockage          |
| Layer L6 recovery / post-event  | 3+2 canopies, independent power, locator, life support                  | Sizing, deployment tests, minimum altitude envelope            | Low-altitude non-availability; canopy failure      |
| Safety Kernel boundary          | Deterministic authorization of safety-critical actuation                | Requirements, formal/robust verification, independence from AI | AI bypass; incomplete preconditions                |

## 36. Failure Containment Philosophy

The architecture identifies unintended separation, partial/asymmetric release, sensor loss, power-bus failure, AI/ML failure, fire during separation, released-component collision, parachute out-of-envelope deployment, landing-support failure, airbag inadvertent/non-deployment, sealed-capsule overpressure/toxicity (OI-18), and seat-stroke bottom-out. Containment philosophy: prevent common-cause defeat of multiple layers; detect degraded states; define a safe fallback for every failed-release configuration; preserve human authority; keep AI advisory at the safety boundary.

| Failure Class                | Containment Design Response                                              | Program Rule                                     |
|------------------------------|--------------------------------------------------------------------------|--------------------------------------------------|
| Unintended separation        | Mechanical locks + electrical inhibits + command validation + monitoring | Quantitative safety objective + CCA              |
| Partial / asymmetric release | State sensing + asymmetric-load analysis + degraded recovery mode        | First-class emergency state, not an afterthought |
| Sensor loss / bad data       | Validity monitoring + diverse sensing + safe-state logic                 | Kernel must fail safe                            |
| Power-bus failure            | Independent survival power for critical functions                        | No single bus defeats L1–L6 critical path        |

| Failure Class                     | Containment Design Response                               | Program Rule                              |
|-----------------------------------|-----------------------------------------------------------|-------------------------------------------|
| AI/ML failure                     | Deterministic Safety Kernel remains authoritative         | Non-negotiable certification boundary     |
| Sealed-capsule airbag gas (OI-18) | Sealed-volume analysis + staged/vented deployment options | Must close before deployment-logic freeze |

## Appendix A — Glossary of Abbreviations

| Abbreviation | Meaning                                                                                        |
|--------------|------------------------------------------------------------------------------------------------|
| CFRP         | Carbon-fibre-reinforced polymer                                                                |
| CG           | Center of gravity                                                                              |
| CCA          | Common-cause analysis                                                                          |
| DOF          | Degrees of freedom (6-DOF: full rigid-body translational + rotational dynamics)                |
| ECU          | Electronic control unit                                                                        |
| FEA          | Finite element analysis                                                                        |
| FM           | Figure of merit (rotor hover efficiency)                                                       |
| FMEA         | Failure modes and effects analysis                                                             |
| FTO          | Freedom to operate                                                                             |
| ICD          | Interface control document                                                                     |
| IP           | Intellectual property                                                                          |
| KE           | Kinetic energy                                                                                 |
| MTOW         | Maximum takeoff weight                                                                         |
| OEI          | One engine inoperative                                                                         |
| OI           | Open item                                                                                      |
| PSSA/SSA/FHA | Preliminary system safety assessment / system safety assessment / functional hazard assessment |
| RVM          | Requirements verification matrix                                                               |
| SFC          | Specific fuel consumption                                                                      |
| TRL          | Technology readiness level (NASA/ESA 1–9 scale)                                                |
| ZSA          | Zonal safety analysis                                                                          |

## Appendix B — Status Label Quick Reference

See Part I, Section 5.1 for the full precise definitions. This is a one-line quick reference for reviewers scanning tables.

| Label                         | One-line meaning                                                |
|-------------------------------|-----------------------------------------------------------------|
| LOCKED LIMIT / LOCKED CONCEPT | Fixed for Generation 1; formal change process required to alter |
| REQ                           | Binding requirement unless formally waived                      |
| TARGET                        | Development objective, not demonstrated                         |
| ASSUMPTION / STUDY            | Working value pending analysis or test                          |
| CALC                          | Calculated from stated assumptions/targets                      |
| OPEN / OPEN — CRITICAL        | Requires closure; critical items sit on the P0 path             |
| DEFERRED — RESEARCH TRACK     | Excluded from the licensable core baseline                      |

**END OF HSH-00 REVISION 23 — PROFESSIONAL MASTER EDITION**

Complete content preserved from Rev 22 FULL | Reorganized into nine parts with working table of contents | Founder:  
Amirbahador Atai  
*Not a certification approval | Not a filed patent | Not demonstrated performance*